Macro-Sensitized Credit Risk Engine: Isolating the 2022 Fed Rate Shock via Causal Inference

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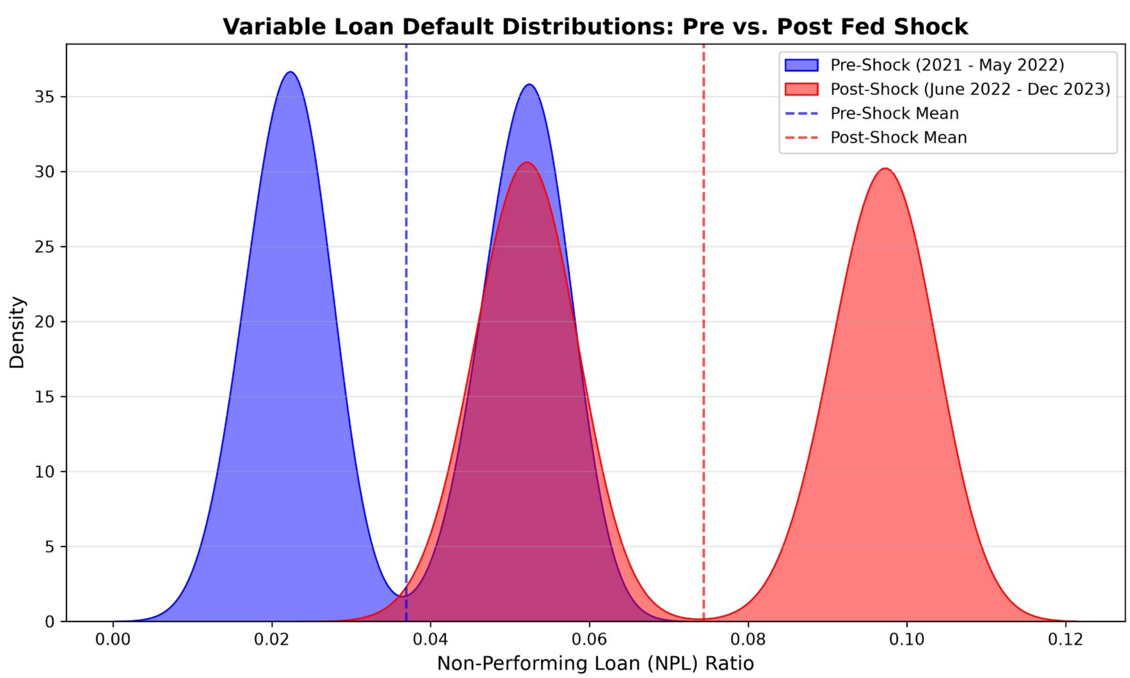
1. Executive Summary & Business Impact

This repository houses a production-grade econometric pipeline designed to dynamically isolate and quantify macroeconomic risks within consumer credit portfolios. The primary objective of this architecture is to provide a scalable engine where proprietary internal loan data can be integrated to instantly generate custom macroeconomic stress-test profiles.

Leveraging a Difference-in-Differences (DiD) 2D Panel Regression framework, this engine moves beyond correlative machine learning to extract true causal parameters while controlling for exogenous macroeconomic noise (e.g., inflation, labor market tightness, and wage growth).

- **Core Validation Finding:** Using simulated cohort data, the model successfully isolated a statistically significant 3.74% causal increase in defaults for variable-rate cohorts directly attributable to the Fed rate shock (p-value: 0.000).



2. Econometric & Methodological Defense

To ensure robust standard errors and prevent false signals, this model strictly adheres to advanced econometric defenses, actively avoiding the pitfalls of standard 1D forecasting.

TEACHING CALLOUT: Why 1D Time-Series Fails (ARIMAX vs. DiD)

- **The Problem:** In 2022, inflation hit 40-year highs and the Fed hiked rates aggressively, yet unemployment remained historically low. A 1D time-series struggles to isolate which of these overlapping variables drove a spike in defaults.
- **The 2D Panel Solution:** By utilizing Fixed-Rate cohorts as a control group (who experienced the exact same inflation and labor market trends but not the rate shock), the model mathematically washes out the macro noise. This isolates the pure, causal impact of the Fed rate hike on the Variable-Rate treatment group.

TEACHING CALLOUT: The Moulton Problem (Idiosyncratic Noise)

- **The Statistical Trap:** Running a macro regression across 10,000 individual loans treats monthly macro data as 400,000 independent data points. This artificially shrinks Standard Errors toward zero, creating false statistical confidence (The Moulton Problem).
- **The Solution:** Loans were aggregated into 144 distinct demographic buckets (e.g., Prime, Fixed-Rate, Jan 2021 Origination). Tracking these cohorts yielded N=3,984 monthly observations. Standard errors in the final OLS model are explicitly clustered at the cohort_id level to align data density with macro reporting frequencies.

The Causal Specification:
Y_it = Beta_0 + Beta_1(Group)_i + Beta_2(Time)_t + Beta_3(Group x Time)_it + Gamma(X)_it + Error_it
(Where Beta_3 represents the isolated causal impact of the rate shock).

3. Software Engineering & ELT Pipeline

The data pipeline relies on a modular ELT (Extract, Load, Transform) architecture designed for enterprise scalability and strict referential integrity.

- **01_sqlite_ingestion.py:** Defensively parses multi-vector macro streams. Utilizes SQLAlchemy ORM for atomic commits, ensuring corrupt or incomplete data batches are automatically rejected by the database to protect model integrity.
- **02_synthetic_loan_generator.py:** Generates the loan lifecycle DGP (Data-Generating Process), injecting random stochastic variance (real-world noise) to validate the regression model's signal-extraction capabilities.
- **03_sql_feature_assembly.py:** Pushes heavy transformations down to the database layer via native CREATE VIEW architecture. This guarantees referential integrity, avoids Pandas in-memory computing bottlenecks, and ensures downstream analysts query a pristine, single-source-of-truth dataset.

- **04_ks_test_eda.py:** Executes non-parametric distribution shift tests (Kolmogorov-Smirnov) to statistically validate pre/post-shock divergences before regression modeling.
- **05_did_regression.py:** Executes the final clustered OLS panel regression via statsmodels.

4. Future Scope & Enterprise Deployment

This engine is designed to be highly modular. Future iterations can easily adapt to different business needs:

- **In-House Data Integration:** The backend schema is mapped to ingest raw internal loan origination and performance files.
- **Public Macro Benchmarks:** Cohort performance vectors can be swapped with public macro-level response variables (e.g., FRED Credit Card Delinquency Rates) and matched against high-frequency leading indicators like Initial Jobless Claims or the VIX to evaluate market-wide distress baselines.

Environment Setup

This project uses standard scientific Python libraries. Ensure dependencies are installed via:

pip install -r requirements.txt